Chapter 3. Probabilistic Regression, Regularization and Kernel Ridge

Linear regression from a probabilistic view, regularization, ridge regression, the kernel trick

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1 Gaussian Likelihood Explains Squared Error

Motivation. Under Gaussian noise, maximum likelihood is least squares; each loss corresponds to a noise model.

Theorem 3.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximizing the likelihood is minimizing the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximizing over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. Sum the logs of the independent densities $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$. In σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

Motivation. Adding $\lambda\|\mathbf{w}\|^2$ makes ill-conditioned least squares well posed and shrinks coefficients.

2.1 Definition and closed form

Definition 3.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 3.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimizer is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$: ridge with $\lambda = \sigma^2/\tau^2$ (small τ, strong shrinkage).

3 Other Penalties and the Lasso

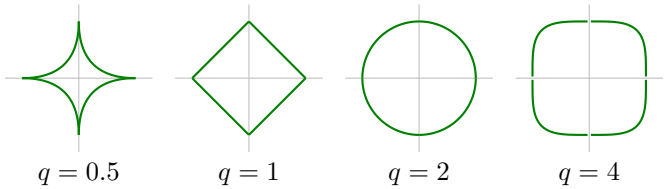
Penalty $\sum_j |\beta_j|^q$: $q = 2$ ridge (pure shrinkage); $q = 1$ lasso (corners on the axes give exact zeros); $0 < q < 1$ nonconvex.

Theorem 3.4: soft thresholding

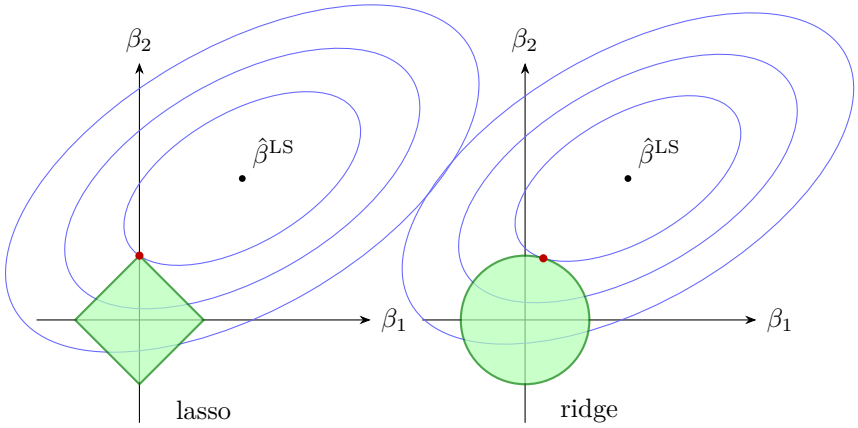
$\arg\min_b \frac{1}{2}(b-z)^2+\lambda|b| = \operatorname{sign}(z)\max(|z|-\lambda,0)$, whereas $\arg\min_b \frac{1}{2}(b-z)^2+\frac{\lambda}{2}b^2 = z/(1+\lambda)$.

Proof. Strict convexity; optimality is $0 \in b-z+\lambda\,\partial|b|$ with $\partial|b| = \{\operatorname{sign} b\}$ $(b \neq 0)$, $[-1,1]$ $(b=0)$. $b>0$: $b = z-\lambda$, iff $z>\lambda$. $b<0$: $b = z+\lambda$, iff $z<-\lambda$. $b=0$: iff $|z| \le \lambda$. Ridge: $b-z+\lambda b = 0$. ■

Lasso shifts coefficients by λ and zeroes small ones; ridge rescales and never zeroes.



Unit balls $\sum_j |\beta_j|^q \le 1$.



The RSS contour meets the lasso diamond at a corner (a zero) and the ridge disk at a generic point.

4 Kernel Ridge Regression

Motivation. Ridge in feature space needs only inner products, so a kernel gives infinitely many features for an $n \times n$ solve.

4.1 Feature-space and sample-space forms

Feature map $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1)\cdots\boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m\times n}$. Ridge $J = \frac{1}{2}\|\mathbf{y}-\Phi^\top\mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$ gives

$$\nabla J = -\Phi\mathbf{y} + (\lambda\mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda\mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 3.5: push-through identity

$(\lambda\mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda\mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i,\mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top\boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K+\lambda\mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top\hat{\mathbf{w}} = \sum_{i=1}^{n}\alpha_i k(\mathbf{x}_i,\mathbf{x}) = \mathbf{k}_{\mathbf{x}}^\top(K+\lambda\mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top \Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ on the left and $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ on the right (both positive definite). ■

Remark 3.6: the kernel trick and cost

Only k is needed. Primal: $m \times m$ system; dual: $n \times n$, better when $m \gg n$ (Gaussian kernel: $m = \infty$). Substituting $\mathbf{w} = \Phi\boldsymbol{\alpha}$ gives $K[(K + \lambda \mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; for singular K the displayed $\boldsymbol{\alpha}$ is one solution and all give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalized: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 3.7: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K \mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

A similarity function is a valid kernel only if all its Gram matrices are PSD.

4.3 Worked examples

Example 3.8: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_\mathbf{x} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 3.9: quadratic kernel

$\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1, 0)$, $\mathbf{k}_{\mathbf{x}_t} = (1, 4)$ and $f(\mathbf{x}_t) = (43 - 8)/69 = 35/69 \approx 0.507$. Check: $\boldsymbol{\phi}(\mathbf{x}_1) = (1, 1, \sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4, 0, 0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1, 0, 0)$ give the same $35/69$.

5 Selected exercises

Motivation. Ridge as augmented least squares, thresholding under orthonormal design, and kernel validity.

Exercise 3.1 (Ridge as augmented least squares; effective degrees of freedom)

Let $X \in \mathbb{R}^{n\times d}$, $X \neq 0$, $\lambda > 0$, and $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

(a) Let $\tilde{X} = \begin{pmatrix} X \\ \sqrt{\lambda}\,\mathbf{I}_d \end{pmatrix}$ and $\tilde{\mathbf{y}} = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}$. Show that $J(\boldsymbol{\beta}) = \frac{1}{2}\|\tilde{\mathbf{y}} - \tilde{X}\boldsymbol{\beta}\|^2$, that $\tilde{X}$ always has full column rank, and recover $\hat{\boldsymbol{\beta}}_\lambda$ from the normal equations of Chapter 1.

(b) The fitted values are $X\hat{\boldsymbol{\beta}}_\lambda = H_\lambda \mathbf{y}$ with $H_\lambda = X(X^\top X + \lambda \mathbf{I})^{-1}X^\top$. Show that $\mathrm{df}(\lambda) := \operatorname{tr} H_\lambda = \sum_{i=1}^{r} \frac{s_i^2}{s_i^2+\lambda}$, where $s_1, \dots, s_r$ are the positive singular values of X.

(c) Show that df is strictly decreasing on $(0, \infty)$, tends to $r = \operatorname{rank} X$ as $\lambda \to 0^+$ and to 0 as $\lambda \to \infty$.

Solution. [Supplement]

(a) Blockwise, $\|\tilde{\mathbf{y}} - \tilde{X}\boldsymbol{\beta}\|^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \|\sqrt{\lambda}\,\boldsymbol{\beta}\|^2 = 2J(\boldsymbol{\beta})$. $\tilde{X}\mathbf{v} = \mathbf{0}$ forces $\sqrt{\lambda}\,\mathbf{v} = \mathbf{0}$: full column rank. With $\tilde{X}^\top\tilde{X} = X^\top X + \lambda\mathbf{I}$, $\tilde{X}^\top\tilde{\mathbf{y}} = X^\top\mathbf{y}$, Chapter 1 gives the unique minimizer $\hat{\boldsymbol{\beta}}_\lambda$.

(b) Full SVD $X = USV^\top$, U, V orthogonal, $S_{ii} = s_i$ $(i \le r)$, zero elsewhere. Then $(X^\top X + \lambda\mathbf{I})^{-1} = V(S^\top S + \lambda\mathbf{I})^{-1}V^\top$ and

$$H_\lambda = US(S^\top S + \lambda \mathbf{I}_d)^{-1}S^\top U^\top.$$

The middle matrix is diagonal with entries $s_i^2/(s_i^2+\lambda)$ $(i \le r)$ and 0; by cyclicity $\operatorname{tr} H_\lambda$ is their sum.

(c) $r \ge 1$ as $X \neq 0$. Each term has derivative $-s_i^2/(s_i^2+\lambda)^2 < 0$ and tends to 1 $(\lambda \to 0^+)$ and 0 $(\lambda \to \infty)$. For full column rank $\mathrm{df}(0^+) = d = \operatorname{tr} H$ (Exercise 1.1). ■

Exercise 3.2 (Orthonormal design: soft, hard and proportional shrinkage)

Assume $X^\top X = \mathbf{I}_d$, let $\lambda > 0$ and $\mathbf{z} = X^\top\mathbf{y}$ (the least squares estimate).

(a) Show that $\frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \frac{1}{2}\|\mathbf{z} - \boldsymbol{\beta}\|^2 + c$ with c independent of $\boldsymbol{\beta}$.

(b) Deduce that the lasso $\arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1$ is $\hat{\beta}_j = \operatorname{sign}(z_j)\max(|z_j| - \lambda, 0)$, and that ridge with penalty $\frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$ gives $\mathbf{z}/(1+\lambda)$.

(c) Show that best-subset selection $\arg\min_{\boldsymbol{\beta}} \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda^2}{2}\|\boldsymbol{\beta}\|_0$, where $\|\boldsymbol{\beta}\|_0$ is the number of nonzero entries, is hard thresholding: $\hat{\beta}_j = z_j$ if $|z_j| > \lambda$ and $\hat{\beta}_j = 0$ if $|z_j| < \lambda$.

Solution. [Supplement]

(a) Using $X^\top X = \mathbf{I}$ and $X^\top\mathbf{y} = \mathbf{z}$,

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \mathbf{y}^\top\mathbf{y} - 2\boldsymbol{\beta}^\top\mathbf{z} + \boldsymbol{\beta}^\top\boldsymbol{\beta} = \|\mathbf{z} - \boldsymbol{\beta}\|^2 + (\mathbf{y}^\top\mathbf{y} - \mathbf{z}^\top\mathbf{z}).$$

(b) By (a) the lasso objective is $c + \sum_j \left[\frac{1}{2}(z_j - \beta_j)^2 + \lambda|\beta_j|\right]$, separable in coordinates; soft thresholding gives $\hat{\beta}_j$, and the same theorem gives ridge $z_j/(1+\lambda)$.

(c) Again $c + \sum_j g_j(\beta_j)$, $g_j(b) = \frac{1}{2}(z_j - b)^2 + \frac{\lambda^2}{2}\mathbf{1}\{b \neq 0\}$: $g_j(0) = z_j^2/2$, and $g_j(b) \ge \lambda^2/2$ for $b \neq 0$ with equality at $b = z_j \neq 0$. If $|z_j| > \lambda$, $\lambda^2/2 < z_j^2/2$ and $b = z_j$ is the unique minimizer; if $|z_j| < \lambda$, $b = 0$. (At $|z_j| = \lambda$ both are optimal.) ■

Exercise 3.3 (Building valid kernels)

Call $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ *valid* if every Gram matrix $K = [k(\mathbf{x}_i, \mathbf{x}_j)]_{i,j=1}^n$ is symmetric positive semidefinite.

(a) Show that if k_1, k_2 are valid and $c \ge 0$, then $k_1 + k_2$ and ck_1 are valid, and so is $k(\mathbf{x}, \mathbf{z}) = g(\mathbf{x})g(\mathbf{z})$ for any $g : \mathcal{X} \to \mathbb{R}$.

(b) Show that if $A, B \in \mathbb{R}^{n\times n}$ are symmetric PSD, so is the entrywise product $A \circ B$; deduce that $k_1 k_2$ is valid.
(c) Deduce that $(1 + \mathbf{x}^\top \mathbf{z})^p$ ($p \in \mathbb{N}$) and the Gaussian kernel $\exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$ are valid on $\mathbb{R}^m$.

Solution. [Supplement]

(a) Symmetry is clear. For $\mathbf{a} \in \mathbb{R}^n$: $\mathbf{a}^\top (K_1 + K_2)\mathbf{a} = \mathbf{a}^\top K_1 \mathbf{a} + \mathbf{a}^\top K_2 \mathbf{a} \geq 0$ and $\mathbf{a}^\top (cK_1)\mathbf{a} = c\,\mathbf{a}^\top K_1 \mathbf{a} \geq 0$. For $k = g(\mathbf{x})g(\mathbf{z})$, $\mathbf{a}^\top K \mathbf{a} = \sum_{i,j} a_i a_j g(\mathbf{x}_i) g(\mathbf{x}_j) = \big(\sum_i a_i g(\mathbf{x}_i)\big)^2 \geq 0$.

(b) By the spectral theorem $B = \sum_k \mu_k \mathbf{q}_k \mathbf{q}_k^\top$ with $\mu_k \geq 0$; put $\mathbf{b}_k = \sqrt{\mu_k}\,\mathbf{q}_k$, so $B_{ij} = \sum_k b_{k,i} b_{k,j}$. Then

$$\mathbf{a}^\top (A \circ B)\mathbf{a} = \sum_k \sum_{i,j} (a_i b_{k,i}) A_{ij} (a_j b_{k,j}) = \sum_k (\mathbf{a} \circ \mathbf{b}_k)^\top A (\mathbf{a} \circ \mathbf{b}_k) \geq 0.$$

$A \circ B$ is symmetric, and the Gram matrix of $k_1 k_2$ is $K_1 \circ K_2$.

(c) *Step 1 (polynomial kernel).* The linear kernel ($\mathbf{a}^\top K \mathbf{a} = \|\sum_i a_i \mathbf{x}_i\|^2$) and the constant 1 ($g \equiv 1$) are valid, so $1 + \mathbf{x}^\top \mathbf{z}$ is valid by (a) and its pth power by (b) and induction.

Step 2 (exponential kernel). $e(\mathbf{x}, \mathbf{z}) = \exp(\mathbf{x}^\top \mathbf{z}/\sigma^2)$ is the entrywise limit of the valid $e_N = \sum_{q=0}^{N} (\mathbf{x}^\top \mathbf{z}/\sigma^2)^q / q!$, so $\mathbf{a}^\top E \mathbf{a} = \lim_N \mathbf{a}^\top E_N \mathbf{a} \geq 0$.

Step 3 (Gaussian kernel). By $\|\mathbf{x} - \mathbf{z}\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{x}^\top \mathbf{z} + \|\mathbf{z}\|^2$,

$$\exp\Big(-\frac{\|\mathbf{x} - \mathbf{z}\|^2}{2\sigma^2}\Big) = g(\mathbf{x})\, e(\mathbf{x}, \mathbf{z})\, g(\mathbf{z}), \qquad g(\mathbf{x}) = \exp\Big(-\frac{\|\mathbf{x}\|^2}{2\sigma^2}\Big),$$

which is valid by (b) applied to e and the kernel $g(\mathbf{x})g(\mathbf{z})$ of (a). ■